Protocol

Efficacy of Simulation-Based Education for intravascular catheterization
: A protocol for systematic review and meta-analysis

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1. Background

The procedure of placing intravenous catheters can cause lethal complications. The ultrasound-guided technique has been the gold standard for central venous catheterization due to its high success rate and low complication rate. However, mechanical complications have been reported to occur even with this technique[1]. Thus, appropriate education is needed to utilize ultrasound guidance, which can lead to the success of catheterization and prevent mechanical complications. In 2014, Madenci et al.[2] showed that simulation training increases the success rate of ultrasound-guided central venous catheterization. Currently, point-of-care ultrasound is popularly use, and many types of vascular access devices are placed using ultrasound guidance. Some RCTs have been done on ultrasound-guided intravascular catheterization. In this paper, we will conduct an updated review of the efficacy of simulation education for ultrasound-guided intravascular catheterization, of which central venous catheters, peripherally inserted central catheters, dialysis catheters, and arterial catheters are used.

2 . Research Question

Does simulation education for ultrasound-guided intravascular catheterization improve the success rate of catheter insertion compared to having no simulation education?

3 . Methods

3.1 Criteria for research to be incorporated into the review

3.1.1 Types of research

We incorporated all randomized controlled trials (RCTs) irrespective of publication status (published and unpublished articles, conference abstracts, and letters), language, and the country in which the study was conducted. We also did not exclude studies based on the length of observation period. Studies conducted with crossover, cluster randomization, or quasi-experimental methods were excluded.

3.1.2. Study participants

To be included in the study, the following criteria were all met: (1) all patients underwent vascular catheterization, and (2) the types of vascular catheters used were either central venous catheters, arterial catheters, peripherally inserted central venous catheters, or dialysis catheters. On the other hand, any studies utilizing catheter replacement were excluded.

3.1.3 Interventions

The placement of vascular catheters was performed by a healthcare provider who has received simulation training (including model and pre-lecture). No exclusions were made based on the experience or occupation of the person performing the puncture. The type of catheter used in the simulation training should be the same as that used in the actual patients. The content of the simulation training was not specified.

3.1.4 Controls

For the control group, we included data from previous vascular catheter insertions done by a healthcare provider without simulation training

3.2 Types of Outcomes

3.2.1 Primary Outcome

The primary outcomes are (1) the success rate, defined as the number of successful punctures divided by the number of punctured patients, and (2) adverse events, as defined by the authors of the individual studies.

3.2.2 Secondary Outcome

The secondary outcomes are (1) the first-time puncture success rate in eligible patients, defined as the pros and cons of success on the first puncture divided by the number of patients, and (2) the number of punctures to success.

3.3 Search methods

3.3.1 Electronic Search

We used the following databases for our search: (1)The Cochrane Central Register of Controlled Trials, (2) MEDLINE via Ovid, (3) EMBASE, (4) ERIC, and (4) CINAHL. The search strategies are listed in Appendix 1, 2, 3, 4, and 5.

3.3.2 Other resources

We further searched ClinicalTrials.gov and ICTRP for ongoing and unpublished studies.

3.4 Data Extraction and Analysis

3.4.1 Selection of the study

Two independent reviewers checked the manuscript titles and abstracts. All of the studies chosen by both reviewers will be subjected to full-text review. Subsequently, the full text was used to determine whether the titles and abstracts can be included in the review independently. If it was an abstract-only study and we were unsure if it met the criteria for review, the original author was contacted. Disagreements between the two reviewers were discussed and resolved. This was discussed with a third reviewer, if necessary.

3.4.2 Data Extraction

Data extraction was performed independently by two reviewers. Disagreements between the two reviewers were discussed and resolved. This was discussed with a third reviewer when deemed necessary. The original author was also contacted if needed. For the data extraction form, a pre-checked form using 10 randomly selected studies was used.

3.5 Bias risk

Two reviewers independently using the Risk of Bias 2 tool[3]. Disagreements between the two reviewers will be discussed, and if this fails, a third reviewer will be acting as an arbiter, if necessary.

3.6 Evaluation of treatment effects

A meta-analysis of relative risk with 95% confidence intervals was conducted for the binary variables. The success rate and first puncture success rate were also evaluated. A meta-analysis of mean difference (MD) and 95% confidence intervals was conducted for the continuous variables, such as the number of punctures to success. The adverse events were summarized according to study-specific definitions, but no meta-analysis has been performed.

3.7 Issues for analysis

For the integration of the means and standard deviations of continuous variables, we followed the methods found in the Cochrane Handbook[4].

3.8 Dealing with missing measurement values

3.8.1 Missing Outcomes

In accordance with the ITT principles. The lack of outcomes is not complementary.

missing statistic
If only the standard error or p-value was reported, the standard deviation is calculated using the Altman[5] method. If it was still unknown even after contacting the author, the standard deviation was calculated from confidence intervals and t-values using the method described in the Cochrane Handbook[4]. Alternatively, it was complemented with a validated method[6]. The validity of these methods was tested in a sensitivity analysis.

3.9 Evaluation of heterogeneity

Visual evaluation was done using a forest plot. The I^2 values were interpreted as follows: 0% to 40% might not be important, 30% to 60% may represent moderate heterogeneity, 50% to 90% may represent substantial heterogeneity, and 75% to 100% represents considerable heterogeneity. If heterogeneity was detected ($I^2 > 50\%$), the cause was verified. For I^2 values, a Cochrane Chi^2 test (Q-test) was performed and a P value < 0.10 was considered statistically significant. Any findings of heterogeneity were addressed in a subgroup analysis.

3.10 Evaluation of Publication Bias
Search clinical trial registry sites (ClinicalTrials.gov, ICTRP) were used for studies that have been completed but not yet published. These were visually evaluated using funnel plots and by performing the Egger test, with P-values < 0.05 considered statistically significant; this was not performed if only 10 or fewer studies were found, or if only studies with similar sample sizes were found.

3.11 Meta-analysis
Review Manager software (RevMan 5.4) was used, following a random-effects model.

3.12 Subgroup Analysis, Search for Heterogeneity
Subgroup analyses were conducted to assess the heterogeneity of clinical study participants and interventions. The first category was the type of catheter, which had the following subgroups: (1) central venous catheter, (2) arterial catheter, (3) peripherally inserted catheter, and (4) dialysis catheter. Next was the classification of puncture site for central venous catheters and dialysis catheters, which had the following subgroups: (1) internal jugular vein, (2) subclavian vein, and (3) femoral vein. Lastly, we also classified studies according to their enforcers: (1) doctors, (2) nurses, and (3) other occupations.

3.13 Sensitivity Analysis

Exclusion of studies using imputed statistics was performed as a sensitivity analysis for the primary outcome.

4. Summary of findings table

We created tables to summarize our findings for the following outcomes based on the recommendations in the Cochrane Handbook[3]: (1) Success rate, (2) adverse events, (3) initial puncture success rate, and (3) number of punctures to success. Each table contains a grading of evidence based on the GRADE approach7[7].

5 Conflict of Interest

The authors declare no conflicts of interest.

6 Funding

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Appendix 1: CENTRAL search strategy

(((((((([mh catheterization] OR [mh catheters]) OR [mh "catheterization, central venous"]) OR [mh "catheterization, peripheral"]) OR CVC:ti,ab) OR "peripherally inserted central catheter":ti,ab) OR PICC:ti,ab) OR "arterial line":ti,ab) AND ((((((([mh "educational status"] OR ([mh "educational status"] OR [mh Education])) OR [mh "simulation training"]) OR [mh "educational measurement"]) OR simulated:ti,ab) OR simulation:ti,ab) OR Education:ti,ab) OR [mh "task performance and analysis"]))

Appendix 2: MEDLINE (via PubMed) search strategy

(((((((("catheterization"[MeSH Terms] OR "catheters"[MeSH Terms]) OR "catheterization, central venous"[MeSH Terms]) OR "catheterization, peripheral"[MeSH Terms]) OR "CVC"[Title/Abstract]) OR "peripherally inserted central catheter"[Title/Abstract]) OR "PICC"[Title/Abstract]) OR "arterial line"[Title/Abstract]) AND ((((((("educational status"[MeSH Terms] OR ("educational status"[MeSH Terms] OR "Education"[MeSH Terms])) OR "simulation training"[MeSH Terms]) OR "educational measurement"[MeSH Terms]) OR "simulated"[Title/Abstract]) OR "simulation"[Title/Abstract]) OR "Education"[Title/Abstract]) OR "task performance and analysis"[MeSH Terms])) AND (((((((("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type]) OR "randomized"[Title/Abstract]) OR "drug therapy"[MeSH Subheading]) OR "placebo"[Title/Abstract]) OR "randomly"[Title/Abstract]) OR "trial"[Title/Abstract]) OR "groups"[Title/Abstract]) NOT ("animals"[MeSH Terms] NOT "humans"[MeSH Terms]))

Appendix 3: EMBASE search strategy

S1 "EMB.EXACT.EXPLODE("catheterization")

S2 "EMB.EXACT.EXPLODE("catheter")

S3 EMB.EXACT("central venous catheterization")

S4 "EMB.EXACT.EXPLODE("central venous catheter")

S5 ab(CVC) OR ti(CVC)

S6 ab(peripherally inserted central catheter) OR ti(peripherally inserted central catheter)

S7 ab(PICC) OR ti(PICC)

S8 ab(arterial line) OR ti(arterial line)

S9 S1 OR S2 OR S4 OR S5 OR S6 OR S7 OR S8

S10 "EMB.EXACT.EXPLODE("educational status")

S11 "EMB.EXACT.EXPLODE("education")

S12 "EMB.EXACT.EXPLODE("simulation training")

S13 "EMB.EXACT.EXPLODE("task performance")

S14 ab(simulated) OR ti(simulated)

S15 ab(simulation) OR ti(simulation)

S16 ab(Education) OR ti(Education)

S17 S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16

S18 (ab(random*) OR ti(random*)) OR (ab(clinical NEAR/1 trial*) OR ti(clinical NEAR/1 trial*)) OR (EMB.EXACT("health care quality"))

S19 S9 AND S17 AND S18

Appendix 4: ERIC

((((((((MH "catheterization+") OR (MH "catheters+")) OR (MH "catheterization, central venous+")) OR (MH "catheterization, peripheral+")) OR TI CVC OR AB CVC) OR TI "peripherally inserted central catheter" OR AB "peripherally inserted central catheter") OR TI PICC OR AB PICC) OR TI "arterial line" OR AB "arterial line") AND ((((((((MH "educational status+") OR ((MH "educational status+") OR (MH "Education+"))) OR (MH "simulation training+")) OR (MH "educational measurement+")) OR TI simulated OR AB simulated) OR TI simulation OR AB simulation) OR TI Education OR AB Education) OR (MH "task performance and analysis+")))

Appendix 5: CINAHL

((((((((MH "catheterization+") OR (MH "catheters+")) OR (MH "catheterization, central venous+")) OR (MH "catheterization, peripheral+")) OR TI CVC OR AB CVC) OR TI "peripherally inserted central catheter" OR AB "peripherally inserted central catheter") OR TI PICC OR AB PICC) OR TI "arterial line" OR AB "arterial line") AND ((((((((MH "educational status+") OR ((MH "educational status+") OR (MH "Education+"))) OR (MH "simulation training+")) OR (MH "educational measurement+")) OR TI simulated OR AB simulated) OR TI simulation OR AB simulation) OR TI Education OR AB Education) OR (MH "task performance and analysis+"))) AND (((MH randomized controlled trials) OR (MH double-blind studies) OR (MH single-blind studies) OR (MH random assignment) OR (MH pretest-posttest design) OR (MH cluster sample) OR (TI (randomised OR randomized)) OR(AB (random*)) OR (TI (trial)) OR (MH (sample size) AND AB (assigned OR allocated OR control)) OR (MH (placebos)) OR(PT (randomized controlled trial)) OR (AB (control W5 group)) OR (MH (crossover design) OR MH (comparative studies))))NOT ((((MH animals+) OR(MH (animal studies)) OR (TI (animal model*))) NOT (MH (human))))